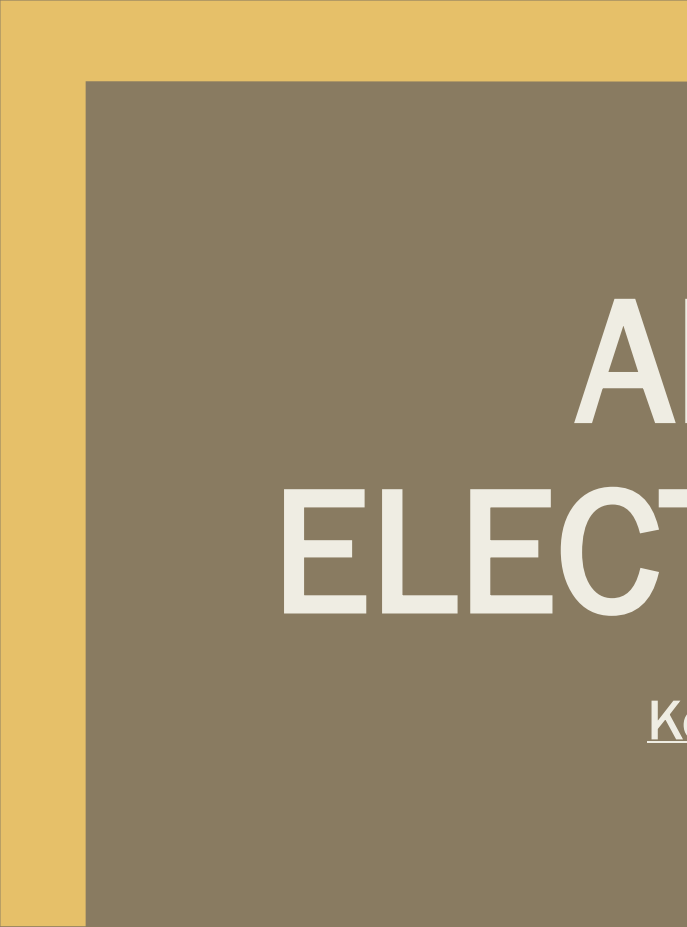




ANTINEUTRON ELECTROPRODUCTION

Kevin Arias • Lei Guo • Leonel Martinez

Florida International University

April 10, 2026

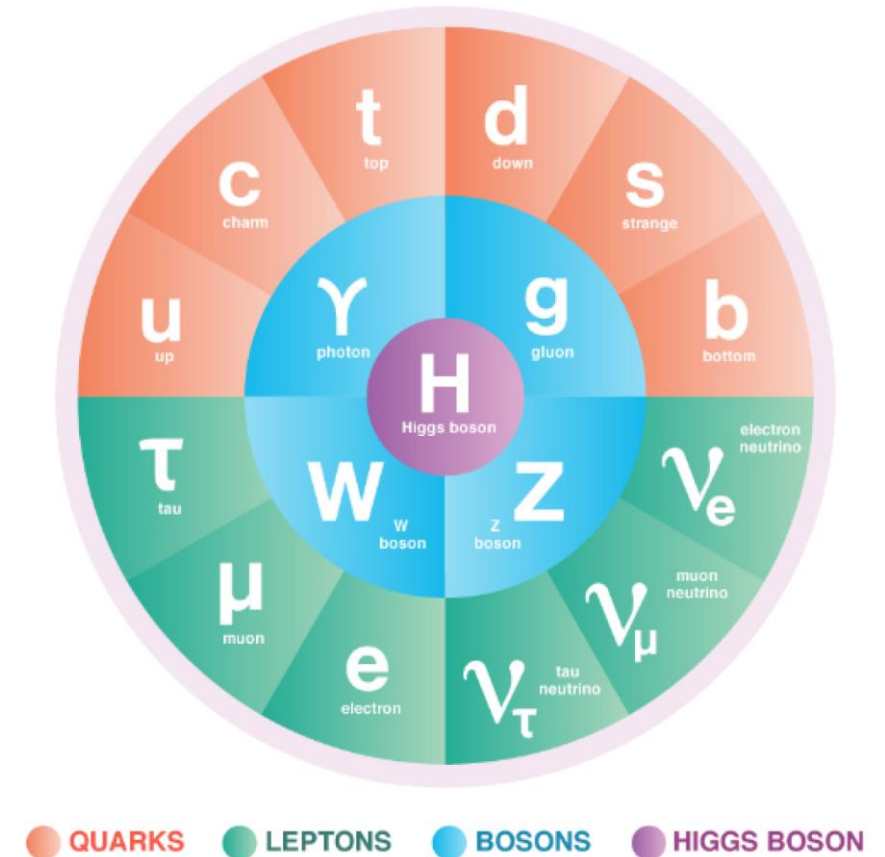


History of Anti-baryons

- Early 20th century scientist thought particle's energy must always be a positive value
- In 1928, Dirac suggested that antiparticles could exist with the same mass as its twin but with reversed sign for electric charge and energy
- In 1955, pp collisions at the Bevatron led to discovery of antiprotons [\[APS, 2016\]](#)
- In 1956, $\bar{p}p$ collisions were used to induce CEX and create “60 antineutrons” [\[Physics Reports, 2003\]](#)
- We know that standard model enforces fundamental conservation laws like conservation of baryon number!

$$\begin{aligned}\bar{p} + p &\rightarrow \bar{n} + n \\ -1 + 1 &\rightarrow -1 + 1\end{aligned}$$

$$\begin{aligned}ep &\rightarrow e'pp\pi^-(\bar{n}) \\ +1 &\rightarrow 1 + 1 \quad (-1)\end{aligned}$$



The Standard Model of Particle Physics
[CERN, 2024]

Background Information

- The Big Bang should have created equal amounts of matter and antimatter, which would have annihilated each other.
- Baryon Asymmetry: A tiny, unexplained imbalance allowed normal matter to survive. Studying antimatter creation helps us probe this cosmic mystery.



Motivation

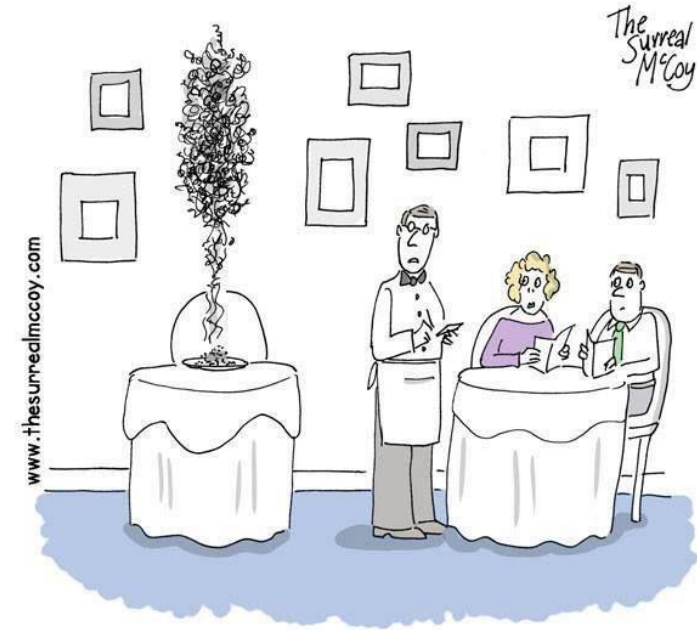
- **The Knowledge Gap:** Relatively little is known about the antineutron's fundamental interactions compared to the antiproton
- **High Statistics:** we can achieve significantly higher statistics than previous historical experiments
- **Future Physics Opportunities:** By successfully creating and clearly isolating antineutrons, we refine our detection methods for the future
- **Opening New Doors:** This paves the way for future experiments that either feature antineutrons in complex final states or potentially utilize antineutron beams.



Positron & Antiproton
Research



Antineutron Research



"It is my understanding that the gentleman ordered both the pasta and the antipasta."

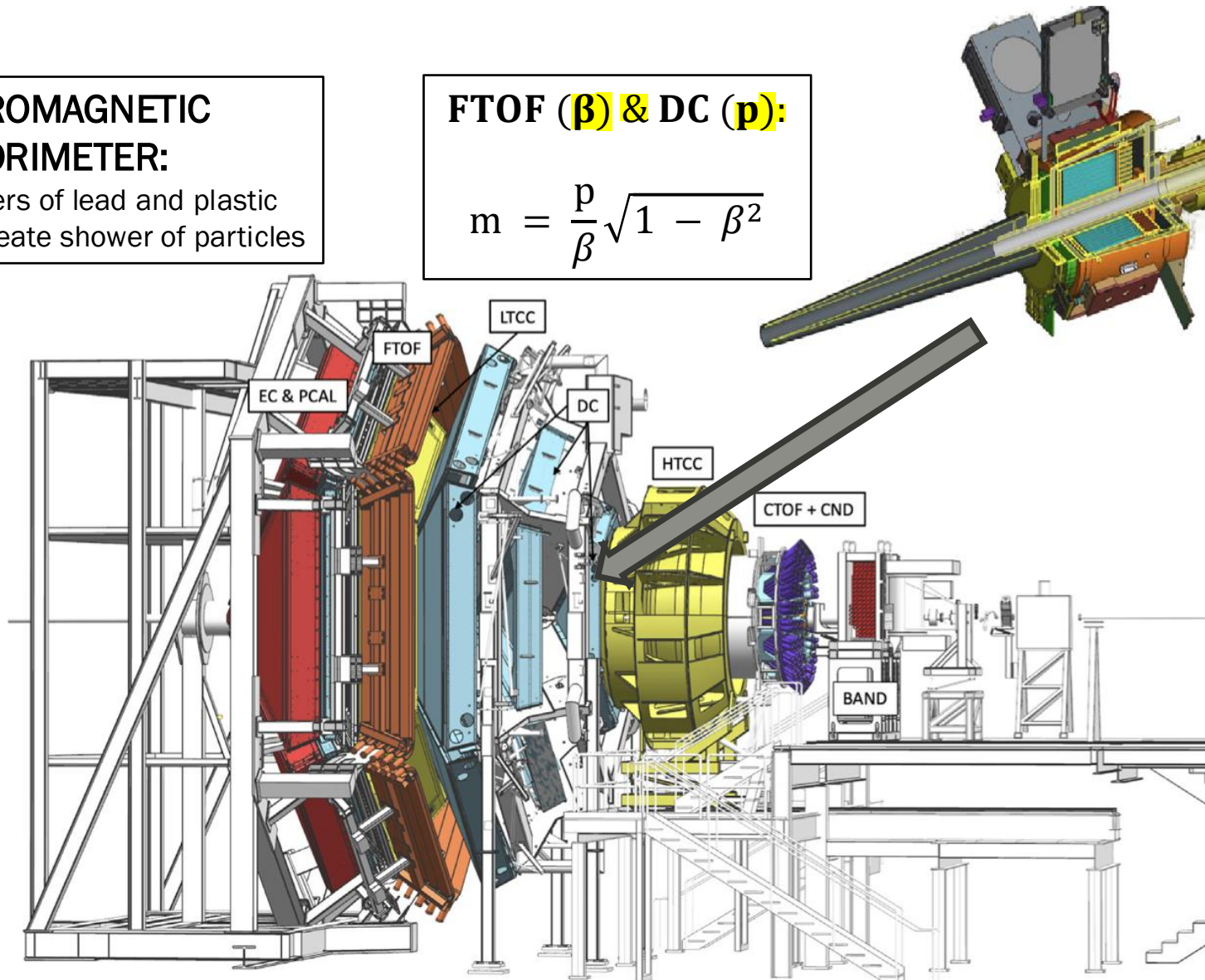
CLAS12 Particle Detector

ELECTROMAGNETIC CALORIMETER:

Alternating layers of lead and plastic
scintillators to create shower of particles

FTOF (β) & DC (p):

$$m = \frac{p}{\beta} \sqrt{1 - \beta^2}$$



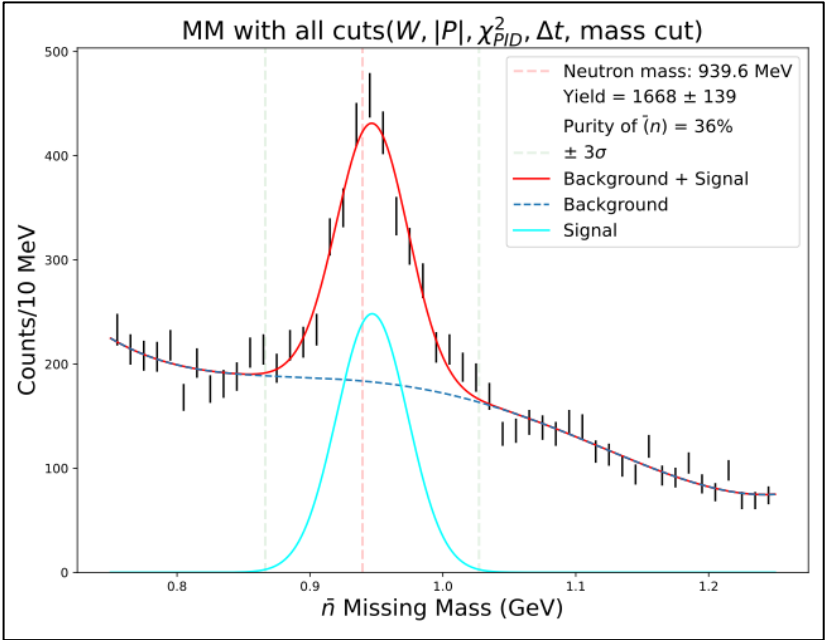
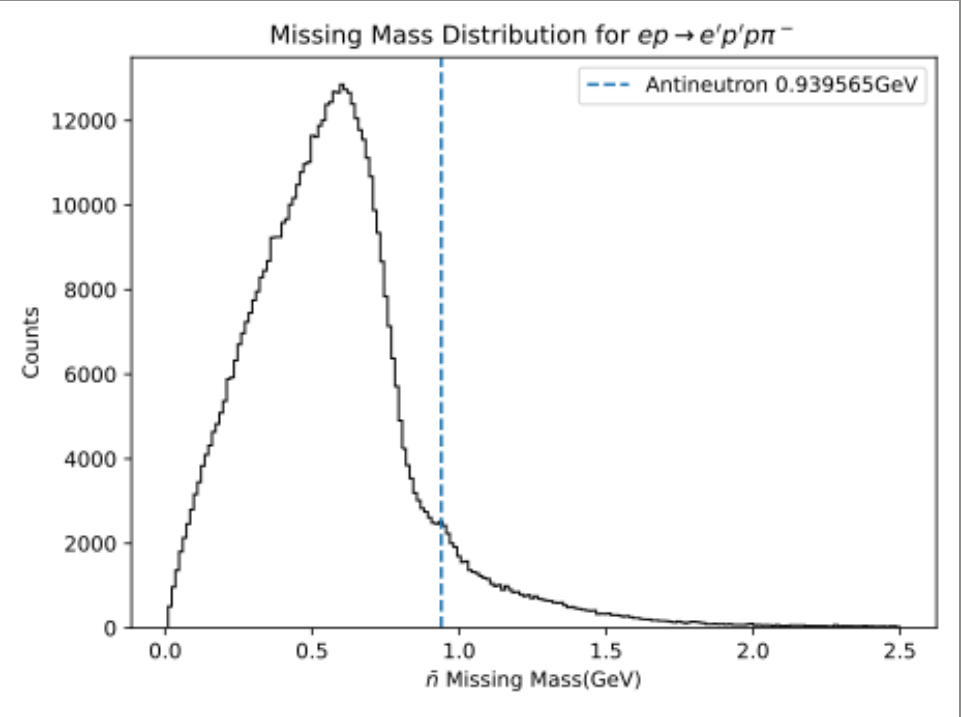
Forward Tracker & Forward ECAL:

Used to capture scattered
electron in $2.5^\circ - 4.5^\circ$ range
and measure

- Thin sensor that confirms
particle has charge $\Rightarrow e$
- Showers from electrons
striking lead-tungstate
crystals are used to
measure exact electron
energy

Cut Optimization (Forward Detector)

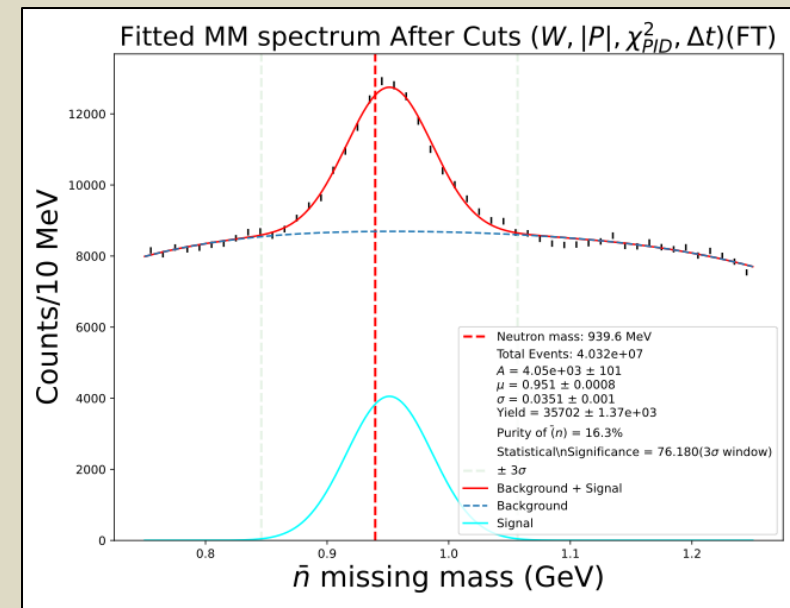
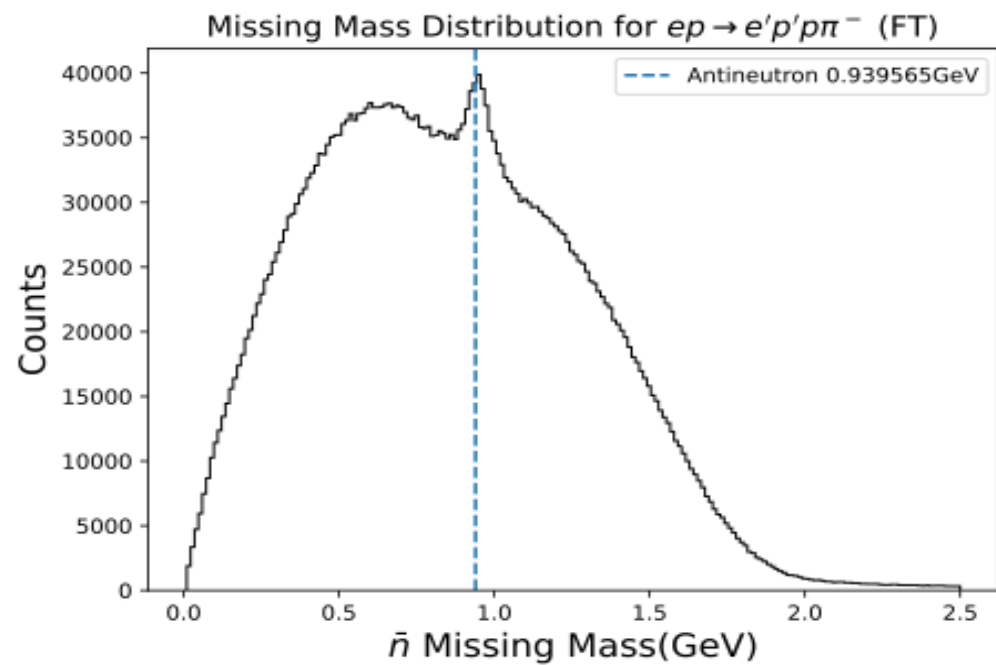
- 1D spectrum of antineutrons is produced using missing mass technique
- Then, apply cuts based on physical variables



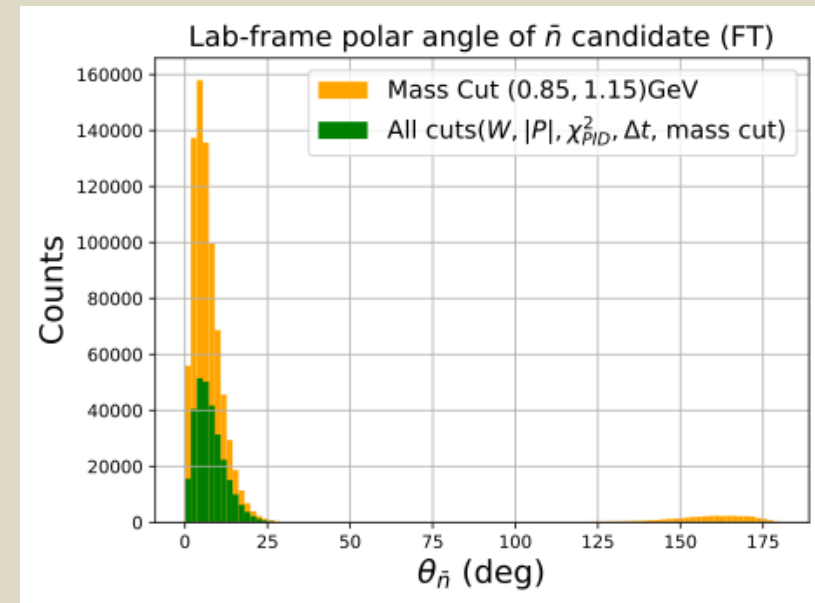
Boolean Mask	Description
Invariant Mass Cut W	Represents total energy available to create new mass
Momentum Magnitude Cuts $ P $	Specific upper and lower bounds for the 3D momentum magnitude of each detected particle
Delta Time Cuts Δt	Time difference between particles' theoretical detector hit and actual detector hit
χ^2_{PID}	Automated detector algorithm that tells you accuracy of identified particle based on how it interacts with detector

Cut Optimization (Forward Tagger)

- **FT:** Catches ultra-low angle scattered electrons (quasi-real photoproduction), yielding exponentially more events

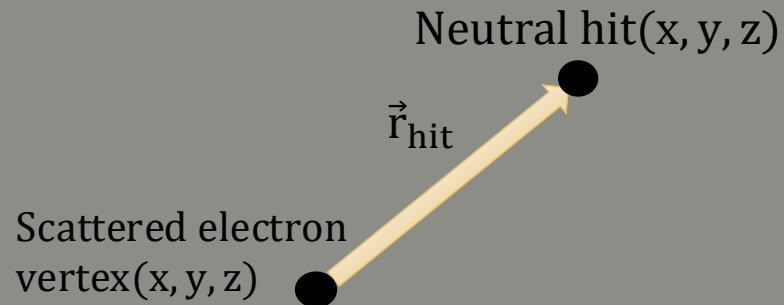


Successful isolation of antineutron and applied fit (Gaussian + 4th order polynomial) allows us to extract the total event yield

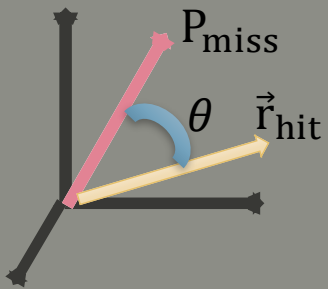


Energy Spectrum of Antineutrons

- By tracking trajectories of known particles (e' , $p1$, $p2$, p_{im}) we can identify untracked energy deposits in ECAL as any neutral candidates ($\bar{n}$, γ)

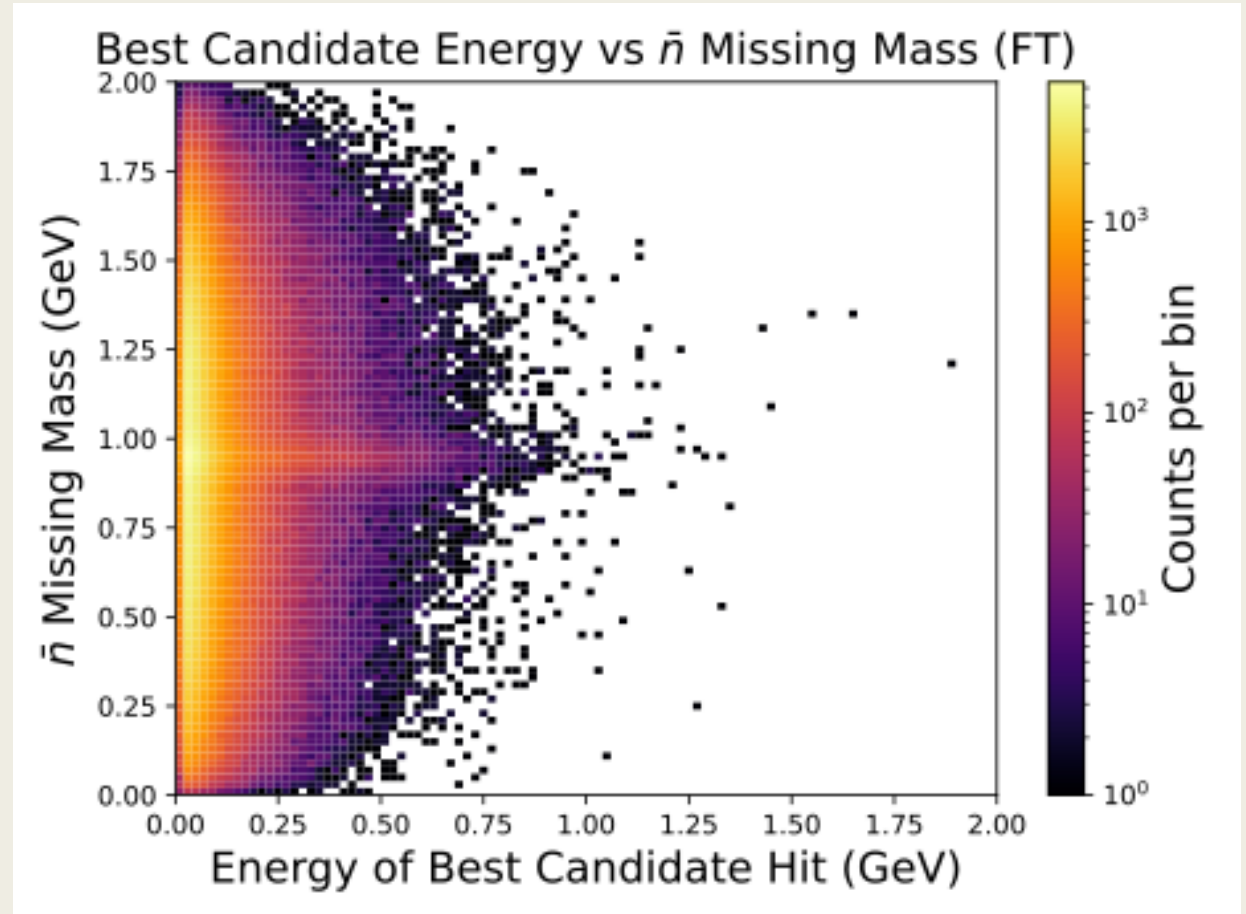


- We calculate the opening angle between the theoretical missing momentum vector and physical location of the neutral hit



$$\frac{\vec{P}_{miss} \cdot \vec{r}_{hit}}{|\vec{P}_{miss}| |\vec{r}_{hit}|} = \cos \theta$$

No previous expectations for energy spectrum of antineutrons in **Electromagnetic** Calorimeter

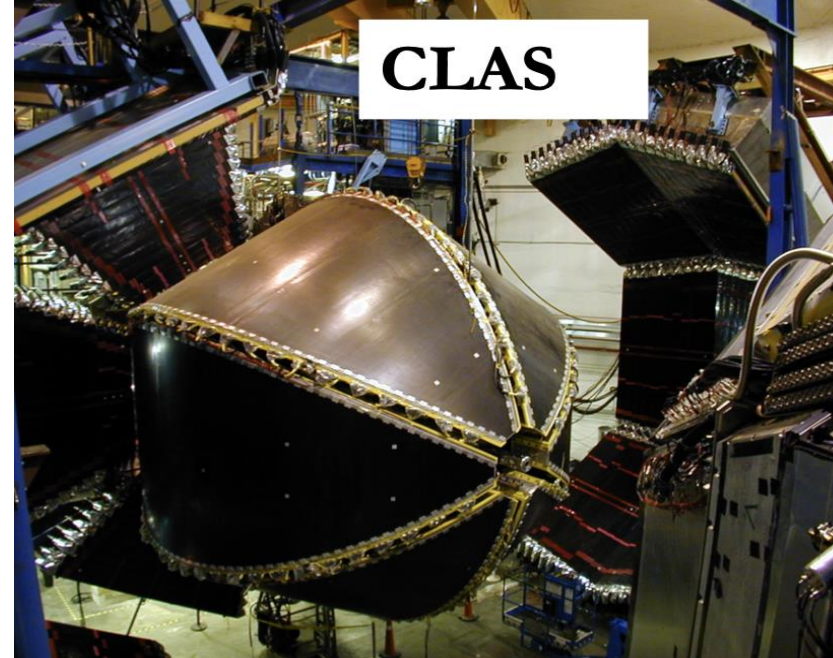
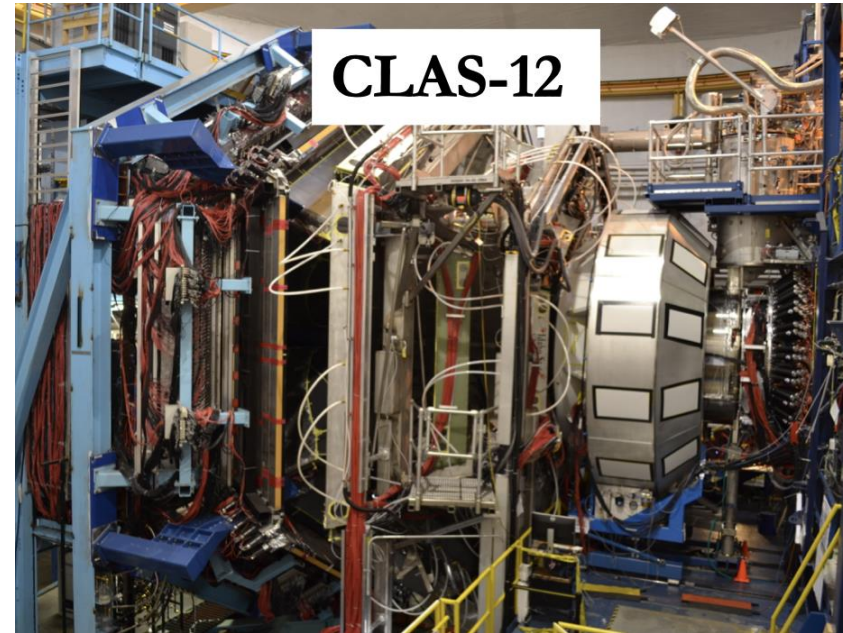


The distinct horizontal band precisely at 0.94 GeV proves our mathematical signal perfectly correlates with physical, neutral energy deposits in the detector.

Conclusion & Outlook

- Compared to photoproduction of antineutrons [\[CLAS 6Gev-12Gev-24Gev upgrade, 2022\]](#)

N	CLAS6(γ) Counts	CLAS12 Counts
# $\bar{p}$	229,117	350-400K
# $\bar{n}$	1,200	35,811



References

CLAS12 Hardware References:

- V. D. Burkert et al., "The CLAS12 Spectrometer at Jefferson Laboratory," *Nuclear Instruments and Methods in Physics Research Section A*, vol. 959, 2020.
- D. Acker et al., "The CLAS12 Forward Tagger," *Nuclear Instruments and Methods in Physics Research Section A*, vol. 959, 2020.

Motivation & Antineutron Scarcity:

- Various Authors, "The unfinished picture of low-energy antineutron interactions: open issues and hints for future research possibilities," *arXiv preprint arXiv:2601.11390*, 2026.
- BESIII Collaboration, "Precision YN and $n\bar{N}$ measurements with an LH2/LD2 target in the BESIII detector," *arXiv preprint arXiv:2603.16075*, 2026.